

1 Solution — GIAM 1.2

Problem 5

1.1 Problem

Complete the following table, related to the conjecture that whenever p is a prime number, $2^p - 1$ is also a prime.

p	$2^p - 1$	prime?	factors
2	3	yes	1 and 3
3	7	yes	1 and 7
5	31	yes	
7	127		
11			

Table 1.1

1.2 The Completed Table

p	$2^p - 1$	prime?	factors
2	3	yes	1 and 3
3	7	yes	1 and 7
5	31	yes	1 and 31
7	127	yes	1 and 127
11	2047	no	1, 23, 89, 2047

Table 1.2

The last row is where the story turns: $2^{11} - 1 = 2047$ is **not** prime.

1.3 Filling In the Last Two Rows

Row $p = 7$. Compute $2^7 - 1 = 128 - 1 = 127$. To test primality, trial-divide by primes up to $\sqrt{127} \approx 11.3$, i.e. by 2, 3, 5, 7, 11:

127 is odd; $1+2+7 = 10$ (not div. by 3); doesn't end in 0/5; $127/7 \approx 18.1$; $127/11 \approx 11.5$

None divides 127, so 127 is prime. Its only factors are 1 and 127.

Row $p = 11$. Compute

$$2^{11} - 1 = 2048 - 1 = 2047.$$

Now trial-divide by primes up to $\sqrt{2047} \approx 45.2$. The first several primes fail:

$$2047/3 \approx 682.3, \quad 2047/7 \approx 292.4, \quad 2047/11 \approx 186.1, \quad 2047/13 \approx 157.5, \quad 2047/17 \approx 120.4$$

but then

$$2047 = 23 \times 89.$$

(Check: $23 \times 89 = 23 \times 90 - 23 = 2070 - 23 = 2047$. ✓) Both 23 and 89 are prime, so the full list of factors of 2047 is 1, 23, 89, 2047 — *four* divisors, not two. Hence 2047 is **composite**.

1.4 The Conjecture Is False

The conjecture " p prime $\Rightarrow 2^p - 1$ prime" holds for the first four primes 2, 3, 5, 7 — and then **fails at $p = 11$** :

$$p = 11 \text{ is prime, yet } 2^{11} - 1 = 2047 = 23 \times 89 \text{ is composite.}$$

So $p = 11$ is a *counterexample*, and the conjecture is disproved. (This is exactly the false "Conjecture 1" flagged in the Section 1.2 reading — a cautionary tale that a pattern holding for several cases is not a proof.)

1.5 Visualizing the Failure

$M = 2^p - 1$ for each prime p : green = M is prime, red = M is composite

p = 2 3 <i>prime</i>	p = 3 7 <i>prime</i>	p = 5 31 <i>prime</i>	p = 7 127 <i>prime</i>	p = 11 2047 <i>23 × 89</i>	p = 13 8191 <i>prime</i>	p = 17 131071 <i>prime</i>	p = 19 524287 <i>prime</i>
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$2^{11} - 1 = 2047 = 23 \times 89$ — the first prime p for which $2^p - 1$ fails

The conjecture works for $p = 2, 3, 5, 7$ — then breaks at 11, even though 13, 17, 19 work again.

Figure 1.1: For $p = 2, 3, 5, 7$ the number $2^p - 1$ is prime (green). At $p = 11$ it is composite ($2047 = 23 \cdot 89$, red). It returns to prime at $p = 13, 17, 19$ — the failures are irregular, not a one-time hiccup.

The picture underlines why the conjecture is so tempting and yet so wrong: it works for the first four primes, breaks at 11, then *works again* at 13, 17, 19. There is no simple pattern to when $2^p - 1$ is prime.

1.6 Remark — These Are Mersenne Numbers

Numbers of the form

$$M_p = 2^p - 1$$

are called **Mersenne numbers**, after Marin Mersenne (1588–1648); when M_p happens to be prime it is a **Mersenne prime**. The conjecture in this problem is the (false) claim that M_p is prime for *every* prime p .

1.7 Remark — The True Statement: Primality of p Is Necessary, Not Sufficient

While " p prime $\Rightarrow M_p$ prime" is false, the *reverse* implication is true and worth knowing:

$$2^p - 1 \text{ prime} \implies p \text{ prime}.$$

Proof (contrapositive). Suppose $p = ab$ is composite with $1 < a, b < p$. Using the factorization $x^b - 1 = (x - 1)(x^{b-1} + \cdots + x + 1)$ with $x = 2^a$:

$$2^p - 1 = 2^{ab} - 1 = (2^a)^b - 1 = (2^a - 1)((2^a)^{b-1} + \cdots + 2^a + 1).$$

Since $1 < a < p$, the factor $2^a - 1$ satisfies $1 < 2^a - 1 < 2^p - 1$ — a *nontrivial* divisor. So $2^p - 1$ is composite. ■

So a prime exponent p is a **necessary** condition for $2^p - 1$ to be prime — which is why the table only tests prime p in the first place. (E.g. $2^4 - 1 = 15 = 3 \times 5$ and $2^6 - 1 = 63 = 7 \times 9$ are composite, predictably, because 4 and 6 are composite.) But it is *not* **sufficient**, as $p = 11$ shows.

1.8 Remark — Concretely, Why 23 Divides 2047

There is a slicker reason than blind trial division. A prime q divides $2^p - 1$ exactly when the order of 2 modulo q divides p ; for prime p this forces the order to be exactly p , which (by Fermat) requires $p \mid (q - 1)$, i.e.

$$q \equiv 1 \pmod{p}.$$

So any prime factor of $2^{11} - 1$ must be $\equiv 1 \pmod{11}$: the candidates are 23, 67, 89, ... (that is, $23 = 2 \cdot 11 + 1$, $67 = 6 \cdot 11 + 1$, $89 = 8 \cdot 11 + 1$, ...). This is why 23 — and not, say, 7 or 13 — is the smallest prime factor: only residue-1-mod-11 primes can possibly divide M_{11} . The search space is tiny, which is how Mersenne factorizations are found in practice.

1.9 Remark — Mersenne Primes Today

Mersenne primes are the headline-grabbers of computational number theory: the **largest known prime** has, for decades, almost always been a Mersenne prime, because the **Lucas–Lehmer test** checks M_p for primality far faster than general numbers of the same size can be tested. The distributed project **GIMPS** (Great Internet Mersenne Prime Search) has found all recent record primes; as of writing, only about 52 Mersenne primes are known, the largest with exponent p in the tens of millions (a number with over 24 million digits).

It is an **open problem** whether there are infinitely many Mersenne primes — and, separately, whether infinitely many M_p are *composite* (this second one is known, in fact, on

heuristic grounds but the distribution is subtle). The irregular green/red pattern in the figure is a small window onto a question nobody can yet fully answer.

1.10 Remark — The Link to Perfect Numbers

Mersenne primes are not just a curiosity — they are in exact correspondence with the **even perfect numbers** (numbers equal to the sum of their proper divisors, like $6 = 1 + 2 + 3$ and $28 = 1 + 2 + 4 + 7 + 14$). The **Euclid–Euler theorem** says:

n is an even perfect number $\iff n = 2^{p-1}(2^p - 1)$ with $2^p - 1$ a Mersenne prime.

So our table is, in disguise, also a table of perfect numbers:

Mersenne prime $2^p - 1$	even perfect number $2^{p-1}(2^p - 1)$
3 ($p = 2$)	$2 \cdot 3 = 6$
7 ($p = 3$)	$4 \cdot 7 = 28$
31 ($p = 5$)	$16 \cdot 31 = 496$
127 ($p = 7$)	$64 \cdot 127 = 8128$

Table 1.3

And because $2^{11} - 1 = 2047$ is *not* prime, $p = 11$ produces *no* perfect number — the failure in our table is also a gap in the list of perfect numbers. (Whether any *odd* perfect number exists is one of the oldest unsolved problems in mathematics.)